Homework 11: Blood Spatter & Glass

Paper track. No home lab. Calculator with arcsin needed.

Part A: Angle of impact (2 pts each)

Compute the angle. `angle = arcsin(width ÷ length)`.

1. Width 3 mm, length 6 mm.
2. Width 4 mm, length 4 mm.
3. Width 2 mm, length 8 mm.
4. Width 6.9 mm, length 8.0 mm.
5. A stain gives an angle of impact of 90°. What did the stain look like, and how did the drop hit?

Part B: Spatter reasoning (2 pts each)

1. A stain is a long teardrop with the point aimed toward the window. Which way was the blood traveling?
2. Match the pattern to the cause: large drips and pools / a fine mist of tiny dots / a straight line of evenly spaced stains across the ceiling.
3. There is a clean, person-shaped blank area in the middle of a spatter pattern on the wall. What is this called, and what does it tell you?
4. You've drawn lines through the long axes of six stains and they all cross near one point. What have you found? What else do you need to find the area of *origin*?

Part C: Glass (2 pts each)

1. Which cracks form first — radial or concentric — and on which side of the force?
2. State the 3R rule.
3. A bullet hole in a window is narrow on the street side and wide on the room side. Was the shot fired from the street or from inside the room?
4. Two bullet holes, X and Y. The radial cracks from Y stop when they reach the cracks from X. Which shot was fired first?
5. Window pane A has a small clean hole with 2 short cracks. Pane B is shattered with a dozen long radial cracks. Which was hit by a high-speed projectile?
